



ME 300: Engineering Thermodynamics

Review of the Course So Far

Structure of the course

1. Defining properties, states, processes $\rightarrow$ assumptions
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2. Energy conservation - 1st Law
3. Limits of energy exchange - 2nd Law

Tool 1: defining properties

Assumptions: continuum, equilibrium

→ properties: macroscopic quantities

→ system, control volume

↳ extensive - M, N, V, U

intensive - T, P, ρ, v, u

Tool 2: defining states

→ state: defined by all properties

→ state relations -

⇒ Assumption: simple compressible substance

$$P_1 = f(P_2, P_3)$$

intensive, independent

Tool 3: conservation stations

↳ Finite time: $\Delta X_{\text{sys}} = X(t=t_2) - X(t=t_1) = X_{\text{in}} - X_{\text{out}} + X_{\text{gen}}$

↳ Instantaneous: $\left. \frac{dX}{dt} \right|_{\text{sys}} = \dot{X}_{\text{in}} - \dot{X}_{\text{out}} + \dot{X}_{\text{gen}}$